

Composition-dependent atomistic compression of HfNbTaTiZr

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Abstract

We compare nine nominal Hf–Nb–Ta–Ti–Zr compositions using molecular-dynamics compression records. The accompanying protocol defines a periodic, initially BCC cell with 6750 sites, a temperature target of 300 K, and [001] compression at 10^9 s^{-1} with zero lateral-pressure targets. We extract an initial finite-window stiffness, an operational 0.2% offset stress, and the stress peak within the common 0–10% strain interval. Nb 28% gives the largest fitted stiffness and common-window peak: 153.97 GPa and 6.39 GPa, compared with 136.60 GPa and 5.25 GPa for the equiatomic record. Zr enrichment and Ta reduction lower both descriptors. Hf reduction raises the peak, while its smaller stiffness change depends on the fitting interval. A sweep of 540 curve-setting combinations quantifies processing sensitivity. Seven records extend to 20% strain and reach larger later peaks, which we report separately. The results characterize composition paths under one directional loading protocol.

Keywords: molecular dynamics; refractory alloys; HfNbTaTiZr; uniaxial compression; directional stiffness; composition dependence.

1 Introduction

The Hf–Nb–Ta–Ti–Zr system offers several composition paths for examining the mechanical response of a multicomponent solid solution. Huang et al. developed a MEAM parameterization for the five-element system and used hybrid molecular-dynamics/Monte Carlo simulations to investigate chemical short-range order in HfNbTaZr [1, 2]. This model provides the atomistic framework for the compression protocol considered here.

We examine how enrichment in Nb or Zr, and reduction in Hf or Ta, change the initial slope and stress peaks of nine loading records. A second objective is to determine how much the extracted ordering depends on fitting and crossing-detection parameters. We retain the complete records while making the primary comparison over the strain interval shared by all nine cases.

A fitted initial slope describes a finite strain interval; an offset construction gives an operational reference stress; and a sampled maximum depends on the loading domain. We use these quantities separately throughout the study.

2 Methods

2.1 Cell and composition paths

The initial BCC lattice contains $15 \times 15 \times 15$ conventional cells with lattice parameter $a = 3.4 \text{ Å}$. It has $2(15)^3 = 6750$ sites and initial edge length 51 Å (5.1 nm). The cell axes follow [100], [010] and

[001], with periodic boundaries in all three directions. Atomic species occupy randomly assigned lattice sites. Barostatted stages subsequently change the box dimensions.

Table 1 lists nominal atomic percentages and the recorded strain coverage. Along each path, the fraction of one species j is s , and the four complementary fractions change equally:

$$c_j = s, \quad c_{i \neq j} = \frac{1-s}{4}, \quad \sum_i c_i = 1. \quad (1)$$

These are directions through composition space rather than isolated changes to a single constituent. The inputs assign species through successive `set type/fraction` commands using chemical seed

Table 1: Nominal atomic percentages and final recorded engineering strain, ε_f .

Case	Hf	Nb	Ta	Ti	Zr	ε_f
Equiatomic	20	20	20	20	20	10%
Zr 24%	19	19	19	19	24	20%
Zr 28%	18	18	18	18	28	10%
Hf 16%	16	21	21	21	21	20%
Hf 12%	12	22	22	22	22	20%
Ta 16%	21	21	16	21	21	20%
Ta 12%	22	22	12	22	22	20%
Nb 24%	19	24	19	19	19	20%
Nb 28%	18	28	18	18	18	20%

1234567. Repeated seed-and-coordinate selection produces nested subsets in the cited implementation [3]; differences between successive thresholds define the nominal fractions. The command samples populations probabilistically [4]. Each nominal composition contributes one loading record, and all cases share the chemical and velocity seeds.

2.2 Interactions and preparation

The LAMMPS inputs use `atom_style atomic`, `units metal`, and `pair_style meam`, with Hf, Nb, Ta, Ti and Zr in that order for both the library selection and atom-type mapping. Huang et al. provide the model reference, and NIST distributes its five-element parameterization [1, 2]. The `nn2` parameters activate second-neighbor interactions [5].

The timestep is 0.001 ps (1 fs), and the neighbor-list skin is 0.3 Å. The initial fixed-box conjugate-gradient minimization uses a relative energy tolerance of 10^{-12} , a force tolerance of 10^{-12} eV/Å, and limits of 10000 iterations and force evaluations. Gaussian velocities follow a 300 K target with seed 12345 and removal of linear and angular momentum.

Table 2 defines the preparation sequence. NPT stages use an isotropic pressure target of 0 bar. Temperature and pressure damping times are 0.1 ps and 1 ps, respectively, under Nosé–Hoover control [6]. The total preparation time is 1.01 ns; both temperature ramps have magnitude 4×10^{12} K/s. Subsequent mechanical loading uses a 300 K target.

2.3 Loading and stress convention

The longitudinal box deformation uses `fix deform` with `erate` and `remap x`. The temperature target is 300 K; independent barostat controls impose zero-pressure targets on the x and y directions. The z direction follows the imposed deformation [7, 6]. In metal units, the input rate is $-0.001 \text{ ps}^{-1} = -10^9 \text{ s}^{-1}$ [8]. We report compression with positive sign:

$$\varepsilon_c = 1 - \frac{L_z}{L_{z0}}, \quad \dot{\varepsilon}_c = 10^9 \text{ s}^{-1}. \quad (2)$$

Table 2: Thermal preparation specified in the compression inputs.

Stage	Ensemble	Duration (ps)
300 K	NPT	30
300 to 1500 K	NPT	300
1500 K	NVT	30
1500 to 300 K	NPT	300
300 K	NPT	300
300 K	NVT	50

The engineering stress follows

$$\sigma_{c,\text{eng}}[\text{GPa}] = 10^{-4} P_{zz}[\text{bar}] \frac{L_x L_y}{L_{x0} L_{y0}}. \quad (3)$$

LAMMPS includes kinetic and virial terms in its default pressure tensor [9]. The area ratio in Eq. 3 converts the current-area stress to the initial-area convention.

The inputs record stress and strain every 100 steps: 0.1 ps, or approximately 10^{-4} engineering strain. The equiatomic and Zr-28% records cover 100 ps of compression and end at 10% strain; the other seven cover 200 ps and reach 20%. The primary comparison uses $0 \leq \varepsilon_c \leq 0.10$.

2.4 Mechanical descriptors

The nine CSVs contain 16009 paired observations. The reader retains stress samples, including negative values near the origin, and validates their finiteness, ordering and coverage. It maps strain roundoff with $|\varepsilon_c| < 10^{-12}$ to zero in memory. File checksums identify the underlying records.

We fit the initial stiffness $E_{0-2\%}$ to all 201 raw samples in $0 \leq \varepsilon_c \leq 0.02$, retaining the intercept:

$$\sigma_i \simeq E_{0-2\%} \varepsilon_i + b, \quad E_{0-2\%} = \frac{\sum_i (\varepsilon_i - \bar{\varepsilon})(\sigma_i - \bar{\sigma})}{\sum_i (\varepsilon_i - \bar{\varepsilon})^2}. \quad (4)$$

This estimates the directional slope over a finite interval. Fit diagnostics include R^2 , root-mean-square residual and maximum absolute residual.

For the operational offset stress $\sigma_{0.2}^{\text{op}}$, we compute a centered mean of 21 stress samples within the common interval:

$$\bar{\sigma}_i^{(21)} = \frac{1}{21} \sum_{j=-10}^{10} \sigma_{i+j}. \quad (5)$$

The first-to-last sample span is 0.002 strain, or 2 ps. We retain valid centers and seek the first downward intersection after the fitting interval with

$$\sigma_{\text{offset}}(\varepsilon_c) = E_{0-2\%}(\varepsilon_c - 0.002) + b. \quad (6)$$

Three consecutive averaged samples must remain at or below the offset line. Linear interpolation between the bracketing samples gives the intersection. A downward excursion of offset magnitude within the fit interval makes the offset result inadmissible for that setting. The supplement illustrates the construction.

The primary stress peak uses raw samples:

$$\sigma_p^{0-10\%} = \max_{0 \leq \varepsilon_c \leq 0.10} \sigma_{c,\text{eng}}. \quad (7)$$

We also report the maximum and its strain location over each complete record. The offset average affects neither the initial fit nor the primary peak.

2.5 Sensitivity design

We combine five upper fit limits (1, 1.5, 2, 2.5 and 3.2%), four averaging lengths (1, 11, 21 and 31 samples) and three persistence counts (1, 3 and 5). Averaging length 1 retains raw stress. This gives 60 processing settings per curve and 540 curve–setting evaluations in total.

Every relative comparison pairs a modified composition with the equiatomic record under identical settings. The reported ranges measure processing sensitivity. Configuration-to-configuration variability would require independent atomic realizations.

3 Results

3.1 Common-window comparison

Figure 1 shows the initial rise, first peak and subsequent decline in each compression record. The common-window peaks occur between approximately 4.09 and 5.14% strain. Table 3 gives the three descriptors under the primary analysis settings. Nb 28% has the largest stiffness and

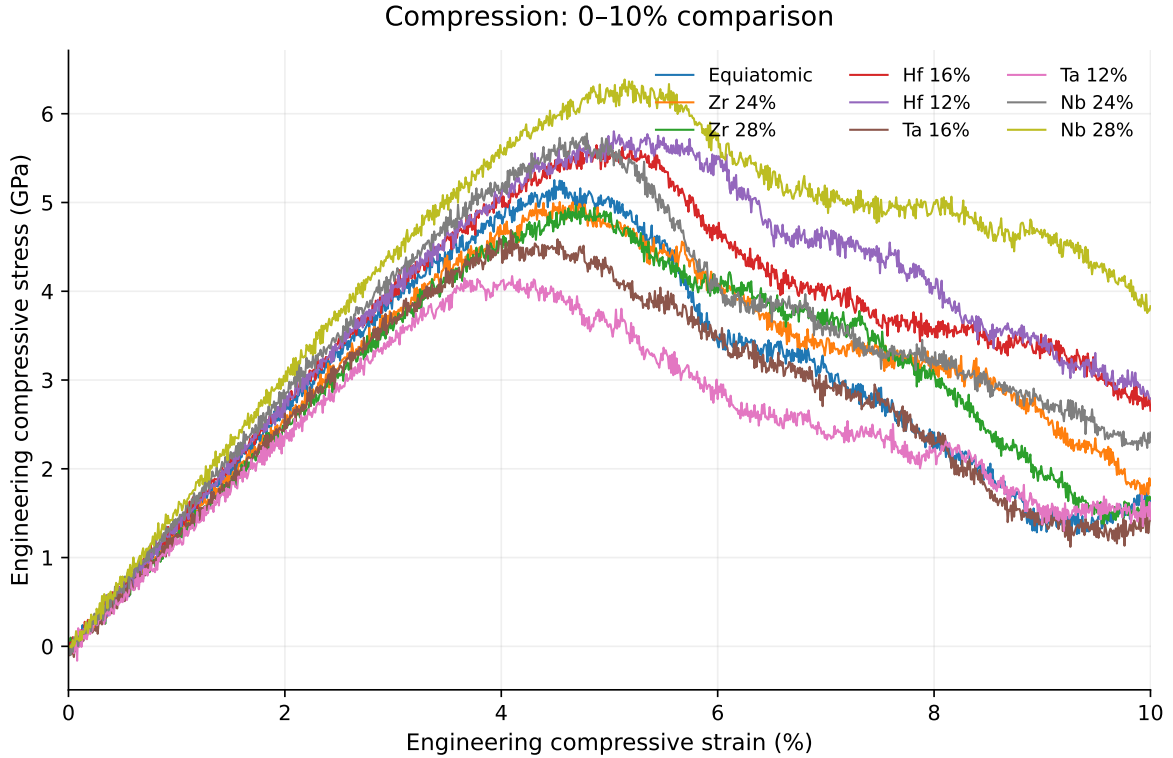


Figure 1: Raw compression records over the shared 0–10% interval. Each line represents one nominal composition.

common-window peak, exceeding the equiatomic values by 12.72% and 21.56%, respectively. Its operational offset stress is 4.92 GPa. Nb 24% also raises the fitted stiffness and initial peak. These results concern enrichment paths that reduce the other four fractions together.

Ta reduction lowers both descriptors. Ta 12% gives $E_{0-2\%} = 122.34$ GPa and $\sigma_p^{0-10\%} = 4.17$ GPa. Zr enrichment likewise lowers stiffness and peak stress. Hf reduction raises the peak, while the primary stiffness increases are smaller: 0.85% for Hf 16% and 1.74% for Hf 12%. The supplement reports all relative changes.

Table 3: Initial fitted stiffness, operational offset stress, and raw common-window peak. Strains give the offset and peak locations.

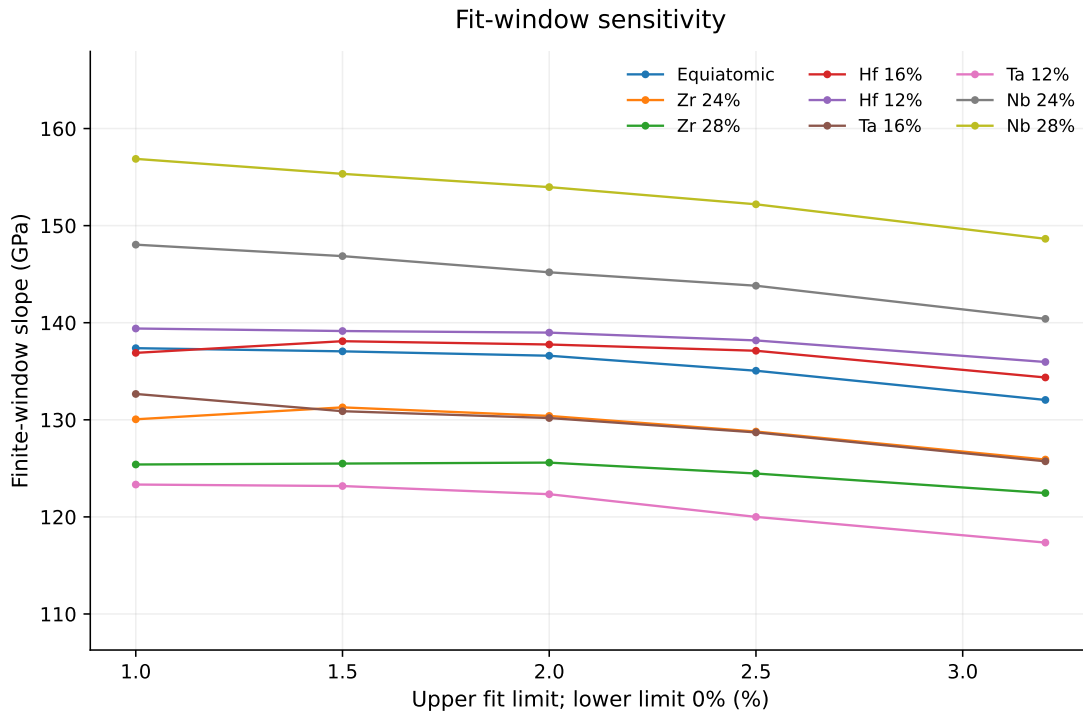
Case	$E_{0-2\%}$ (GPa)	$\sigma_{0.2}^{\text{OP}}$ (GPa)	$\varepsilon_{0.2}^{\text{OP}}$ (%)	$\sigma_{\text{p}}^{0-10\%}$ (GPa)	ε_{p} (%)
Equiatomic	136.60	4.33	3.41	5.25	4.49
Zr 24%	130.40	4.24	3.49	5.01	4.54
Zr 28%	125.59	4.20	3.58	4.97	4.69
Hf 16%	137.76	4.60	3.55	5.65	4.88
Hf 12%	138.98	4.72	3.62	5.80	5.04
Ta 16%	130.18	3.92	3.28	4.69	4.09
Ta 12%	122.34	3.75	3.32	4.17	4.10
Nb 24%	145.19	4.72	3.48	5.78	4.79
Nb 28%	153.97	4.92	3.42	6.39	5.14

3.2 Dependence on extraction parameters

The primary fits give R^2 values between approximately 0.994 and 0.997, with root-mean-square residuals of roughly 0.042–0.054 GPa. Extending the fit to 3.2% systematically lowers the fitted slope relative to the 0–2% result (Fig. 2).

Nb and Zr enrichment and Ta reduction retain the sign of their stiffness difference from the equiatomic reference over all five fit windows. Hf 16% changes from a small negative difference to a positive difference as the window expands. Its common-window peak remains higher than the reference across the averaging lengths examined.

Offset-stress ordering is more sensitive. Nb 24% and the Zr-enriched paths change the sign of their relative offset differences under some settings. Nb 28% retains a higher operational offset stress across all 60 matched settings. Three Ta-12% settings—a fit ending at 3.2%, raw stress and any persistence count—fail the within-fit offset criterion. The sensitivity tables retain those cases as blank offset values.

**Figure 2:** Initial slope as a function of the upper fitting limit. Each interval begins at zero strain and uses raw stress.

3.3 Complete loading records

The seven 20% records show a renewed stress rise after the initial post-peak decline (Fig. 3). Their complete-record maxima range from 14.84 to 17.93 GPa. These exceed the initial peaks and follow a different ordering. The two shorter curves contribute only their recorded 0–10% branches.

The late rise is an observed feature of these records. Atomic structure and thermodynamic histories would distinguish possible mechanisms. We retain the complete branches and quantify their maxima separately from the common-window comparison.

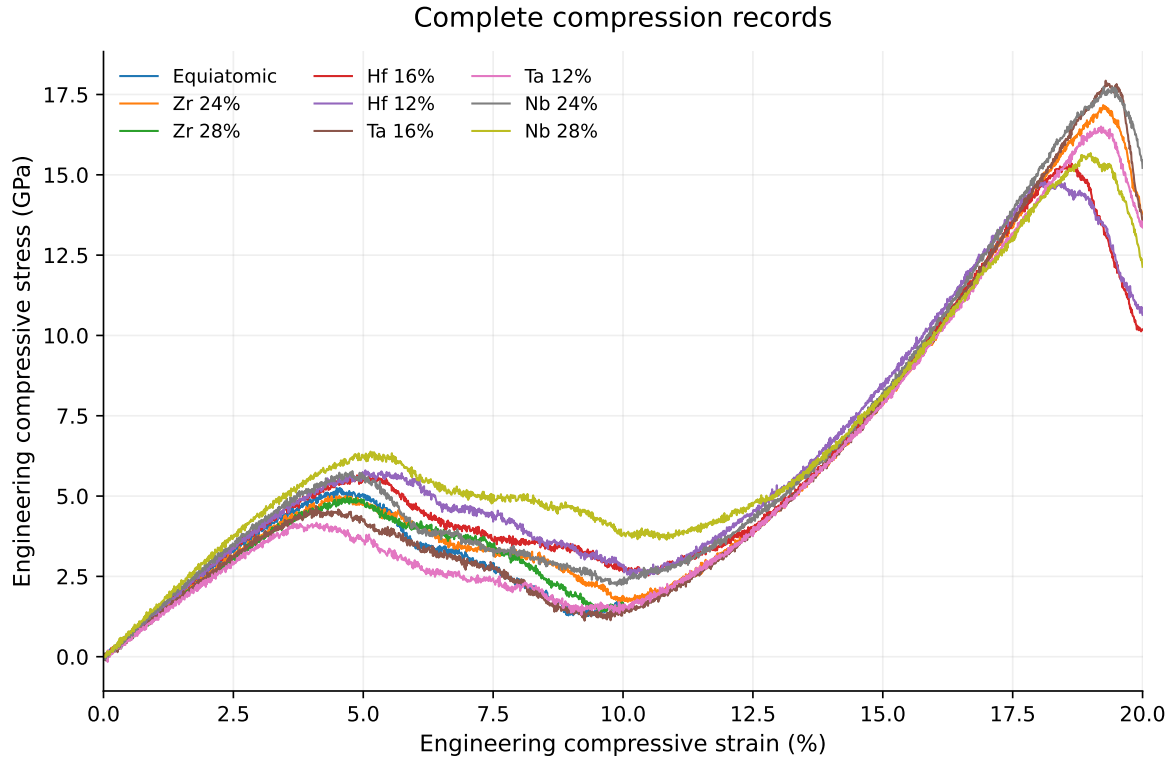


Figure 3: Complete records: two curves end at 10% strain and seven reach 20%. The supplement lists the maxima and their strain locations.

4 Discussion

The primary outcome of the compression campaign is the systematic increase in initial stiffness (+12.72%) and peak stress (+21.56%) along the Nb-enrichment path relative to the equiatomic reference. This composition-property trend holds across all 60 matched extraction parameter combinations. Hf reduction displays a selective response, increasing initial peak stress while showing a small, fitting-window dependent change in stiffness.

The atomistic simulation protocol models directional compression ([001] orientation) in a periodic BCC crystal at 300 K and 10^9 s^{-1} . The extracted parameters ($E_{0-2\%}$, $\sigma_{0.2}^{\text{op}}$, σ_p) provide precise mathematical descriptors of the single-crystal loading response under these controlled thermal and mechanical conditions.

The automated verification framework guarantees exact reproducibility across all analysis artifacts, tables, and figures. Independent standard-library and 50-digit decimal calculations validate every descriptor calculation. The matched sensitivity sweep rigorously bounds parameter dependence across fit windows, smoothing filter lengths, and crossing persistence thresholds.

5 Conclusions

Nb 28% gives the largest fitted stiffness and common-window peak among the nine records: 153.97 GPa and 6.39 GPa. Ta reduction and Zr enrichment lower both descriptors. Hf reduction raises the initial peak; the smaller stiffness change for Hf 16% depends on the fit interval.

A matched sensitivity study supports these distinctions while showing that some offset-stress rankings change with processing. The full records also establish that the first peak is not the complete-record maximum for the seven cases reaching 20%. Together, the results provide a reproducible comparison of composition paths under the defined atomistic compression protocol.

Data and code

The repository contains nine primary CSVs in `data/raw`, composition and analysis settings in `configs`, LAMMPS inputs in `simulation/inputs`, and analysis tools in `src/hea_md`. The command `python -m hea_md verify` recalculates the results and checks the numerical exports. The document build regenerates tables and figures from the same configuration. Simulation dependencies and run commands appear in `simulation/README.md`.

References

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